

Annotation-Free, High-Fidelity SAR Oil-Spill Image Synthesis via Classification-Guided Diffusion Model

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1 Introduction

Marine oil-spill detection is a critical task for environmental protection and maritime safety. Synthetic Aperture Radar (SAR) imagery is widely used for this purpose because of its capability to operate independently of daylight and weather conditions. Oil spills appear as dark regions in SAR images due to the damping of capillary waves on the sea surface.

Recent advances in deep learning have significantly improved oil-spill detection accuracy. However, these methods require large volumes of pixel-level annotated SAR data, which are extremely scarce due to the infrequency of oil-spill events, high acquisition costs, and labor-intensive manual labeling. Existing solutions such as transfer learning and meta-learning alleviate data scarcity to some extent but do not expand the underlying data distribution.

Generative Adversarial Networks (GANs) have been explored for synthetic SAR image generation. Despite their success, GAN-based methods suffer from training instability, mode collapse, and limited control over generated image semantics, often producing blurred or unrealistic oil-spill patterns.

Diffusion models provide a robust alternative by gradually denoising random noise to generate images, offering stable training and high-fidelity outputs. However, existing diffusion-based SAR synthesis methods do not generate pixel-level oil-spill annotations. This work addresses this gap by proposing a Classification-Guided Diffusion Model (CG-DM) capable of generating realistic SAR oil-spill images together with their corresponding segmentation masks in an annotation-free manner.

2 Theoretical Background

Diffusion models generate images through a forward process that incrementally adds Gaussian noise to training samples and a reverse process that learns to remove this noise step by step. Unlike GANs, diffusion models optimize a well-defined likelihood objective, which leads to more stable training and improved diversity in generated samples.

However, conventional diffusion models lack task-specific guidance and semantic control. For SAR oil-spill generation, this limitation results in insufficient morphological accuracy and poor alignment with real spill patterns. Therefore, additional conditioning mechanisms are required to guide the diffusion process toward oil-spill-specific structures and textures.

The proposed CG-DM enhances the standard diffusion framework by integrating classification guidance, label synchronization, and spatial-aware attention. These components inject domain knowledge related to oil-spill morphology and SAR backscatter characteristics directly into the denoising trajectory.

3 Label-Synchronized Generation

The label-synchronized generation module enables simultaneous synthesis of SAR images and their corresponding pixel-level oil-spill masks. This design eliminates the need for manual annotation, significantly reducing labeling cost and human error.

A dual-branch encoder-decoder architecture is employed. The encoder is based on a ResNet-34 backbone combined with Spatial Pyramid Pooling (SPP) to capture multiscale morphological features. The decoder follows a U-Net structure with skip connections and includes a dedicated branch for label generation.

During the reverse diffusion process, the noisy image and its segmentation mask are concatenated into a joint embedding space. Cross-attention mechanisms inject label information into the image generation pathway, ensuring semantic consistency between the synthesized image and mask.

Training is supervised using a pixel-wise L1 loss for image reconstruction and a Dice loss for segmentation accuracy. This dual-supervision strategy ensures that every generated SAR oil-spill image is inherently accompanied by a precise annotation mask.

4 Morphology-Aware Classification Guidance

Oil spills exhibit diverse geometric characteristics in SAR imagery. To model this variability, the proposed framework categorizes oil spills into three dominant morphological types: blocky aggregations, elongated slicks, and patchy distributions.

The training dataset is partitioned into these categories using K-means clustering based on shape descriptors such as area ratio and aspect ratio. Category-specific embeddings are then injected into each reverse diffusion step, guiding the denoising process toward the desired morphological class.

This classification-guided strategy enables precise geometric control over generated oil spills while preserving intra-class diversity. As a result, the generated samples more accurately reflect the statistical distribution of real-world oil-spill morphologies.

5 Spatial-Aware Attention Mechanism

Standard self-attention mechanisms compute global dependencies across the entire feature map, which often leads to oversmoothing and loss of local detail. This limitation is particularly problematic for SAR oil-spill imagery, where fine boundary structures and speckle textures are crucial.

To address this issue, a spatial-aware attention mechanism is introduced. The feature map is divided into non-overlapping local windows, and self-attention is computed independently within each window. This localized attention preserves fine-grained spatial details while reducing computational complexity.

By focusing on localized regions such as oil–water boundaries, the spatial-aware attention mechanism enhances edge sharpness, texture fidelity, and structural realism in the synthesized SAR images. The mechanism is embedded across multiple diffusion stages to progressively refine spatial details during denoising.

6 Dataset Description

Experiments are conducted using the publicly available M4D Oil Spill Dataset derived from Sentinel-1 SAR imagery. The dataset includes VV-polarized Ground Range Detected (GRD) products acquired between 2015 and 2017 with a spatial resolution of 10 meters.

All images undergo radiometric calibration and speckle filtering before being cropped into 256×256 pixel patches. Only oil-spill and non-oil-spill labels are retained for this study. The dataset captures a wide range of environmental conditions, enabling the diffusion model to implicitly learn SAR backscatter variations caused by wind and sea-surface dynamics.

7 Evaluation Metrics

To objectively assess the quality of generated images, three standard metrics are employed: Inception Score (IS), Frechet Inception Distance (FID), and Kernel Inception Distance (KID).

The Inception Score evaluates both perceptual quality and inter-class diversity of generated samples. FID measures the distance between feature distributions of real and generated images under a Gaussian assumption. KID computes the Maximum Mean Discrepancy without assuming Gaussianity, making it more robust for limited datasets.

Lower FID and KID values combined with a higher IS indicate superior image quality, diversity, and distributional consistency.

8 Methodology

The proposed Classification-Guided Diffusion Model (CG-DM) follows a data-driven diffusion framework enhanced with task-specific guidance. The methodology consists of three core components: label-synchronized generation, morphology-aware classification guidance, and spatial-aware attention.

During training, image-mask pairs are used to jointly optimize image reconstruction and segmentation accuracy. Morphological category embeddings are injected into every reverse diffusion step to guide geometric structure formation. Spatial-aware attention refines local textures and boundaries throughout the denoising process.

The entire framework is trained end-to-end, enabling the diffusion model to learn complex SAR backscatter patterns and oil-spill morphology directly from data without requiring explicit physical parameterization.

9 Performance Analysis

Quantitative evaluations demonstrate that CG-DM significantly outperforms state-of-the-art GAN and diffusion-based baselines. The model achieves a Frechet Inception Distance (FID) of 203.14 and a Kernel Inception Distance (KID) of 0.124, indicating high fidelity and strong distributional alignment with real SAR imagery.

Ablation studies confirm the effectiveness of each component. Removing the spatial-aware attention mechanism results in degraded edge sharpness and increased distributional divergence. Classification guidance substantially improves morphological consistency across generated samples.

Furthermore, downstream oil-spill detection experiments reveal that augmenting limited real datasets with CG-DM synthetic samples improves Intersection over Union (IoU) by up to 81.8%. These results highlight the practical utility of CG-DM for real-world maritime monitoring under extreme data scarcity.